

THE GMAT®
QUANT CHEATSHEET



PEMDAS: The orders of mathematical operations

Parentheses

Exponents

Multiplication & Division

Addition & Subtraction

I. Properties of numbers

Primes numbers less than 100

2, 3, 5, 7, 11, 13, 17, 19, 23, 29,
31, 37, 41, 43, 47, 53, 59, 61, 67,
71, 73, 79, 83, 89, 97

→ 1 is not a prime number

→ The only even prime number is 2

→ There are 25 prime numbers under 100

Even/Odd rules : Addition & Subtraction

- $Odd + Odd = Even$
- $Even + Even = Even$
- $Even + Odd = Odd$
- $Odd - Odd = Even$
- $Even - Even = Even$

Even/Odd rules : Multiplication

- $Even \times Even = Even$
- $Even \times Odd = Even$
- $Odd \times Even = Even$
- $Odd \times Odd = Odd$

Even/Odd rules : Division

- $\frac{Even}{Odd} = Even$
- $\frac{Odd}{Odd} = Odd$
- $\frac{Even}{Odd} = Even \text{ or } Odd$

Factors

If y divides evenly into x, we say y is a factor of x

Example:
What are the factors of 12 ?

→ 1, 2, 3, 6, 12

Finding the factors in a particular number: The multiplication method

• Step 1:
Write the given number as the product of two numbers in different possible ways.

• Step 2:
All the numbers that are involved in all these products are the factors of the given number.

Example:
What are the factors of 24 ?

$$\begin{aligned} 24 &= 1 \times 24 \\ 24 &= 2 \times 12 \\ 24 &= 3 \times 8 \\ 24 &= 4 \times 6 \end{aligned}$$

→ 1, 2, 3, 4, 6, 8, 12 and 24, are factors of 24

Finding the number of factors in a particular number

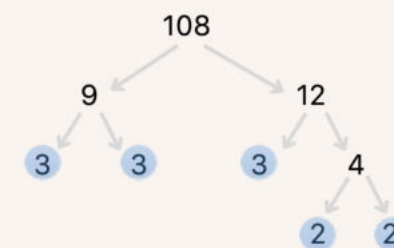
• Step 1:
Find its prime factorisation

• Step 2:
Write the prime factorisation in the exponent form

• Step 3:
Add 1 to each exponent

• Step 4:
Multiply all the resultant numbers. This product is the number of factors

Example:
Find the number of factors in 108 ?



• Step 1 & 2:
 $108 = 3 \times 3 \times 3 \times 2 \times 2$
 $108 = 3^3 \times 2^2$

• Step 3:
 $3+1 = 4$
 $2+1 = 3$

• Step 4:
 $4 \times 3 = 12$

Thus, the number of factors of 108 is 12.



Finding the Greatest Common Factor (the GCF)

Example:
What is the GCF of 56 and 60 ?

$$56 = 2 \times 2 \times 2 \times 7$$

$$70 = 2 \times 5 \times 7$$

$$\rightarrow \text{GCF} = 2 \times 7$$

$$= 14$$

What is the GCF of 132, 198, and 330 ?

$$132 = 2 \times 2 \times 3 \times 11$$

$$198 = 2 \times 3 \times 3 \times 11$$

$$330 = 2 \times 3 \times 5 \times 11$$

$$\rightarrow \text{GCF} = 2 \times 3 \times 11$$

$$= 66$$

Two consecutive integers

Two consecutive integers will never share the same prime factors. Thus the **GCF of two consecutive integer is 1.**

Multiples

A multiple of a number is the product of that number and any integer

Finding the Least Common Multiple (the LCM)

- Step 1:
Find the prime factorization of each integer.
- Step 2:
Of any repeated prime factors among the integers in the set, take only those with the largest exponent. If we're left with two of the same power (for example, 3^2 and 3^2), just take that number once
- Step 3:
Of what is left, take all non repeated prime factors of the integers.
- Step 4:
Multiply together the resultants of steps 2 and 3. The result is the LCM

Example:
What is the LCM of 12 and 56 ?

$$12 = 2 \times 2 \times 3$$

$$56 = 2 \times 2 \times 2 \times 7$$

$$\text{LCM} = 2 \times 2 \times 3 \times 2 \times 7$$

$$= 168$$

What is the LCM of 18 and 42 ?

$$18 = 2 \times 3 \times 3$$

$$42 = 2 \times 3 \times 7$$

$$\text{LCM} = 2 \times 3 \times 3 \times 7$$

$$= 126$$

LCM $\times$ GCF

(LCM of x and y)(GCF of x and y) = xy

Division: the Gmat formula

$$\frac{x}{y} = \text{Quotient} + \frac{\text{remainder}}{y}$$

Example:

$$\frac{22}{5} = 4 + \frac{2}{5}$$

Divisibility rules

- Number divisible by 2
if the ones digit is **0, 2, 4, 6, or 8**
- Number divisible by 3
if the **sum** of all the digit is divisible by **3**
- Number divisible by 4
If the **last two digits** of a number are a number divisible by **4**
- Number divisible by 5
If the **last digit** is **0, or 5**
- Number divisible by 6
If the number is divisible by **both 2, and 3**
- Number divisible by 8
If the **last three digits** of a number are a number divisible by **8**
- Number divisible by 9
If the **sum** of all the digits is divisible by **9**



•Number divisible by 11

If the **sum** of the **odd-numbered place digits** minus the **sum of the even-numbered place digits** is divisible by 11.

Any Factorial $\geq 5!$

Any factorial $\geq 5!$ Will always have zero as its units digit.

Trailing zeroes

The number of trailing zeros of a number is the number of (5×2) pairs in the prime factorization of that number.

Examples:

→ 520 can be expressed as $52 \times 10 = 52 \times (5 \times 2)$ and thus has **one** trailing zero

→ 5200 can be expressed as $52 \times 100 = 52 \times 10^3 = 52 \times (5 \times 2)^2$ and thus has two trailing zeros

Leading zeroes in a Decimal

If X is an integer with k digits, then $\frac{1}{X}$ will have k-1 leading zeroes unless x is a perfect power of 10, in which case there will be k-2 leading zeroes

Terminating Decimals

The decimal equivalent of a fraction will terminate if and only if the denominator of the reduced fraction that contains only 2s or 5s, or both.

Examples:

$$\frac{1}{20} = 0,05$$

$$\rightarrow \frac{1}{12} = 0,08333333...$$

Pattern in Units digits

•Number 0:

All powers of 0 end in 0.

•Number 2:

The unit digits of positive powers of 2 will follow the four- number patterns **2-4-8-6**

•Number 3:

The unit digits of power of 3 will follow
The four number patterns **3-9-7-1**

•Number 4:

The unit digits of power of 4 will follow
The two number patterns **4-6**

•Number 5:

All positive integer powers of 5 end in 5.

•Number 6:

All positive integer powers of 6 end in 6.

•Number 7:

The unit digits of powers of 7 will follow the four number patterns **7-9-3-1**

•Number 8:

The unit digits of positive powers of 8 will follow the four-number patterns **7-9-3-1**

•Number 9:

The unit digits of power of 9 will follow
The two number patterns **9-1**

II. Roots & Exponents

Perfect Squares to know by heart

$0^2 = 0$	$1^2 = 1$	$2^2 = 4$
$3^2 = 9$	$4^2 = 16$	$5^2 = 25$
$6^2 = 36$	$7^2 = 49$	$8^2 = 64$
$9^2 = 81$	$10^2 = 100$	$11^2 = 121$
$12^2 = 144$	$13^2 = 169$	$14^2 = 196$
$15^2 = 225$		

Exponents to Memorize

Base of 2:

$2^0 = 1$	$2^1 = 2$	$2^2 = 4$
$2^3 = 8$	$2^4 = 16$	$2^5 = 32$
$2^6 = 64$	$2^7 = 128$	$2^8 = 256$
$2^9 = 512$	$2^{10} = 1024$	

Base of 3:

$3^1 = 3$	$3^2 = 9$	$3^3 = 27$
$3^4 = 81$	$3^5 = 243$	

Base of 4:

$4^1 = 4$	$4^2 = 16$	$4^3 = 64$
$4^4 = 256$		

Base of 5:

$5^1 = 5$	$5^2 = 25$	$5^3 = 125$
$5^4 = 625$		

Squaring integers ending in 5:

→ Let n be the product before the 5, write the product of n and n+1, followed by 25

Examples:
What is 65^2 ?

$$65^2$$

$$6 \times 7 = 42$$

$$\rightarrow 65^2 = 4225$$

What is 115^2 ?

$$115^2$$

$$11 \times 12 = 142$$

$$\rightarrow 115^2 = 14225$$

Perfect Cubes to know by heart

$0^3 = 0$	$1^3 = 1$	$2^3 = 8$
$3^3 = 27$	$4^3 = 64$	$5^3 = 125$
$6^3 = 216$	$7^3 = 343$	$8^3 = 512$
$9^3 = 729$	$10^3 = 1000$	

Non-Perfect square roots to know by heart

$\sqrt{2} \approx 1,4$	$\sqrt{3} \approx 1,7$	$\sqrt{5} \approx 2,2$
------------------------	------------------------	------------------------

Multiplying Radicals

$$\sqrt[m]{a} \times \sqrt[m]{b} = \sqrt[m]{ab}$$

Example:

$$\sqrt{5} \times \sqrt{3} = \sqrt{15}$$

Dividing Radicals

$$\sqrt[m]{a} \times \sqrt[m]{b} = \sqrt[m]{ab}$$

$$\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$$

Example:

$$\frac{\sqrt{48}}{\sqrt{6}} = \sqrt{\frac{48}{6}} = \sqrt{8} = 2\sqrt{2}$$

Addition and subtraction of Radicals

$$\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}$$

Example:

$$\sqrt{16} + \sqrt{4} \neq \sqrt{20}$$

$$\sqrt{16} + \sqrt{4} = 4 + 2 = 6$$

Multiplication of like bases

$$(x^a) \times (x^b) = x^{a+b}$$

Division of like bases

$$\frac{x^a}{x^b} = x^{a-b}$$

Power to a power rule

$$(x^a)^b = x^{ab}$$



Multiplication of Different bases and like Exponents

$$(x^a) \times (y^a) = (xy)^a$$

Division of Different bases and like exponents

$$\frac{x^a}{y^a} = \left(\frac{x}{y}\right)^a$$

Radical expressed in exponential form

$$\sqrt[b]{a} = x^{\frac{a}{b}}$$

Example:

$$\sqrt{x} = x^{\frac{1}{2}}, \sqrt[3]{x} = x^{\frac{1}{3}}$$

Raising a base to a negative exponent :

$$x^{-1} = \frac{1}{x}, \text{ and in general}$$

$$x^{-y} = \frac{1}{x^y}$$

Examples:

$$3^{-2} = \frac{1}{3^2}, \quad \frac{1}{5^6} = 5^{-6}$$

$$\left(\frac{5}{2}\right)^{-3} = \left(\frac{2}{5}\right)^3$$

III. Word problems

Percent problems

• **A is what percent of B ?**

$$\frac{A}{B} \times 100 = ?$$

• **«Percent less than»**
Translations

$$Final = \left(1 - \frac{\% \text{ Less than}}{100}\right) \times (Initial)$$

Example:

X is 60% less than y

$$\rightarrow X = 0,4y$$

• **«Percent greater than»**
Translations

$$Final = \left(1 + \frac{\% \text{ Greater than}}{100}\right) \times (Initial)$$

Example:

X is 60% greater than y

$$\rightarrow X = 1,6y$$

Percent change formula

$$\left(\frac{Finalvalue - Initialvalue}{Initialvalue}\right) \times 100$$

Simple interest

Simple interest = Principal $\times$ Rate $\times$ Time

Compound interest

$$\rightarrow A = P\left(1 + \frac{r}{n}\right)^{nt}$$

A= amount of interest

P= principal

r= rate of interest

n=number of times per year

interest is compounded

t= time in years

Motion/Distance problem

General formula:

my Ferrari goes up to 300km (distance) per hour (time) !

$$Rate = \frac{Distance}{Time}$$

$$Distance = Rate \times Time$$

• **Motion in opposite direction:**

$$d_1 + d_2 = \text{Total distance}$$

• **Motion in the same direction:**

$$d_1 = d_2$$

• **Round trip:**

$$d_1 = d_2$$

Work problems

General formula:

$$Rate \times Time = Work$$

$$Time = \frac{Work}{Rate}$$

$$Rate = \frac{Work}{Time}$$

Example:

Tom can write 20 pages in 4 hours. His rate is:



$$\text{Rate} = \frac{\text{Work}}{\text{Time}}$$

$$\text{Tom's Rate} = \frac{20\text{pages}}{4\text{hours}} = \frac{5\text{pages}}{1\text{hour}}$$

A machine can produce 15 helmets in 1 week. Its rate is:

$$\rightarrow \frac{15\text{helmets}}{\text{week}}$$

Combined worker formula

Work object 1 + **Work** object 2 = **Work** total

Ratios

$$\frac{\text{number of cats}}{\text{number of dogs}} = \frac{3 + m}{m}$$

This is not a ratio

$$\frac{\text{number of cats}}{\text{number of dogs}} = \frac{5m}{m} = \frac{5}{1}$$

This is a ratio

Combine ratios with the LCM:

$$\text{Ratio 1:} \\ \frac{x}{y} = \frac{3}{4}$$

$$\text{Ratio 2:} \\ \frac{x}{z} = \frac{7}{11}$$

We need to find the LCM of x to combine the two ratios: the LCM of 3 and 7 is 21

Hence,

$$x/y/z = 21/28/33$$

IV. Probability

Complementary events

$$\rightarrow P(A) + P(\text{not } A) = 1$$

The probability of A and B

- if A and B are independent

$$\rightarrow P(A \text{ and } B) = P(A) \times P(B)$$

- if A and B are not independent

$$\rightarrow P(A \text{ and } B) = P(A) \times P(B|A)$$

The addition rule

- Mutually exclusive events

$$\rightarrow P(A \text{ or } B) = P(A + B)$$

- Not mutually exclusive events

$$\rightarrow P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

V. Statistics

Average (Arithmetic Mean)

$$\text{Average} = \frac{\text{Sum of terms}}{\text{Number of terms}}$$

Median

When a set is numerically ordered, the median is the value in the middle of the arranged set

Calculating median with an Odd numbers of terms

Where n is the number of terms in the set

$$\text{Median} = \frac{(n + 1)}{2} \text{ position}$$

Calculating median with an Even numbers of terms

Median = average of the values at the $\frac{n}{2}$ position and the $\frac{n + 2}{2}$ position

Mode

The mode is the number that appears most frequently in a data set

Range

= Highest number in a set - Lowest number in a set

Set of Consecutive integers

Counting the Numbers of Integers in a set of Consecutive integers (inclusive)

Highest number - Lowest number + 1

Counting the Numbers of Multiples of an integer in a set of Consecutive integers (inclusive)

$$\left(\frac{\text{Highest multiple} - \text{Lowest multiple}}{\text{given number}} \right) + 1$$

Average (Arithmetic mean in a set of consecutive integers)

$$\frac{\text{Highest number} - \text{Lowest number}}{2}$$



Evenly spaced sets

Examples:

{4,5,6,7,8}

{3,5,7,9,11}

{12,24,36,48,60}

→ In any evenly spaced set
Mean=Median

Standard deviation: important rules

• Adding/subtracting a constant to each number in a set

→ **The standard deviation does not change**

• Multiplying/ dividing each term in a set of numbers by a constant

→ **The standard deviation will also be multiplied or divided by that amount.**

VI. Enumeration

Order **does** not matter:
Combination

$$\rightarrow {}_nC_k = \frac{n!}{k!(n-k)!}$$

n= number of object in the set
k= number of object selected

Order **does** matter:
Permutation

$$\rightarrow {}_nP_k = \frac{n!}{(n-k)!}$$

n= number of object in the set
k= number of object selected

Mississippi rule

Given n object where A are alike, another B are alike, another C are alike and so on, the number of ways to arrange the n objects is

$$\rightarrow \frac{n!}{(A!)(B!)(C!) \dots}$$

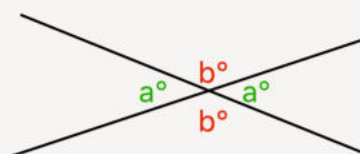
Number of ways to arrange a set of items in a circle

$$\rightarrow (k-1)!$$

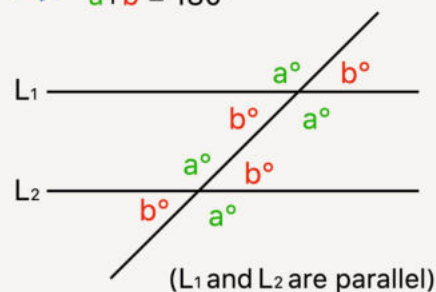
k= number of objects to be arranged in the circle

VII. Geometry

Angles



$$\rightarrow a + b = 180^\circ$$



→ Any **acute** angle + any **obtuse** angle = 180°

Triangles:

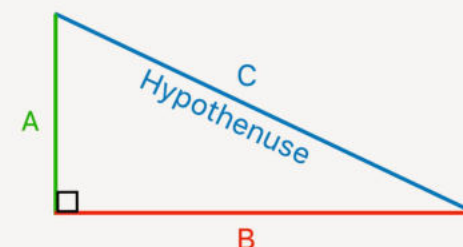
Area of a triangle

$$\text{Area} = \frac{\text{base} \times \text{height}}{2} = \frac{1}{2}bh$$

The triangle inequality theorem

Difference between other 2 sides < 3rd side < sum of other 2 sides

Pythagorean theorem



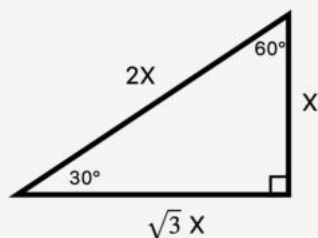
$$A^2 + B^2 = C^2$$

Pythagorean triples (& their multiples)

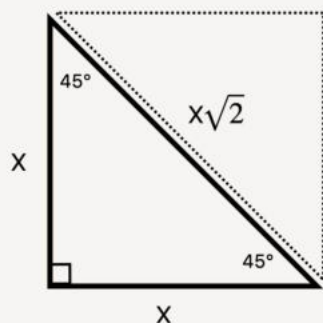
- 3-4-5
- 5-12-13
- 8-15-17 ...

Ratios of special triangles

- The 30-60-90 right triangle



- The half a square triangle



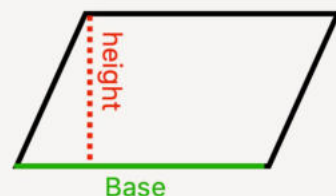
The area of an equilateral triangle

$$\rightarrow \frac{\text{side}^2 \times \sqrt{3}}{4}$$

Longest line in a rectangle

$$\rightarrow \text{Diagonal} = \sqrt{\text{length}^2 + \text{width}^2}$$

The parallelogram



$$\rightarrow \text{Area} = \frac{\text{base} \times \text{height}}{2}$$

- Opposites are equal
→ Opposites angles are equal

Interior angles of a polygon

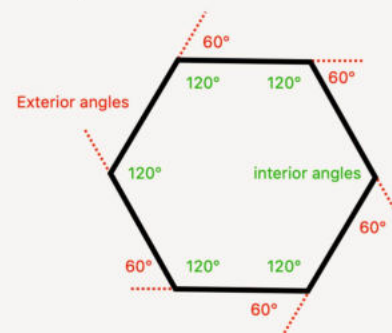
In a n sides polygon, the sum of the interior angles is

$$\rightarrow (n-2) \times 180$$

Exterior angles of a polygon

- They always sum up to 360°

Example:



$$\begin{aligned} \rightarrow 6 \times 60 &= 360 \\ \rightarrow (6-2) \times 180 &= 720 \end{aligned}$$

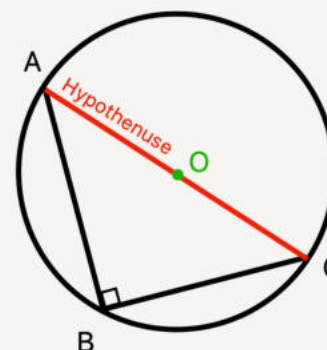
Circles

r = radius

$$\text{Area} = \pi r^2$$

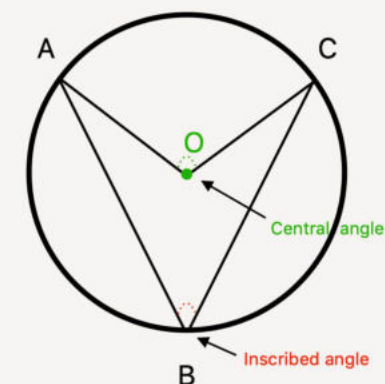
$$\text{Circumference} = 2\pi r$$

Right triangle inscribed in a circle



AC is the hypotenuse of the right triangle and the diameter of the circle

Inscribed Angles in Circles

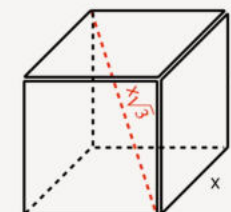


The central angle = 2 inscribed angles

$$\rightarrow \angle AOC = 2\angle ABC$$

The Diagonal of a cube

$$\text{Diagonal} = \text{side} \sqrt{3}$$



VIII. Coordinate Geometry

Slope of a line

$$\rightarrow \text{Slope} = m = \frac{Y_2 - Y_1}{X_2 - X_1}$$

With:

Y_2 = the second y-coordinate

Y_1 = the first y-coordinate

X_2 = the second x coordinate

X_1 = the first x coordinate

m = the slope of the line

The slope-intercept equation

$$\rightarrow Y = mx + b$$

With:

Y = the y-coordinate

x = the x-coordinate

m = the slope of the line

b = the y intercept of the line

Parallel lines

Parallel lines have the same slope but different y-intercept

Perpendicular lines

l_1 and l_2 are perpendicular

m_1 and m_2 are their respective slopes

$$\rightarrow \text{Then } m_1 = -m_2$$

Reflections

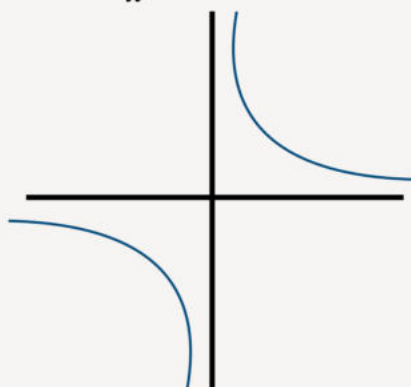
- Reflection over the x axis
 $(x, y) \rightarrow (x, -y)$

- Reflection over the y axis
 $(x, y) \rightarrow (-x, y)$

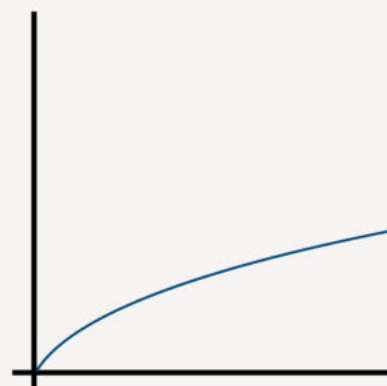
- Reflection over the origin
 $(x, y) \rightarrow (-x, -y)$

Graphic representations of functions to have in mind

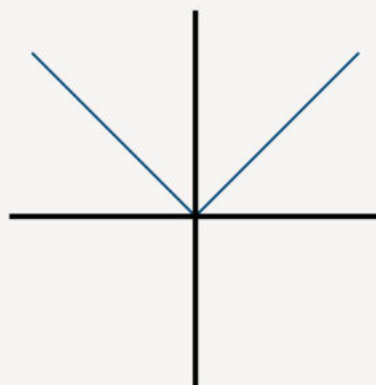
$$\bullet x \rightarrow \frac{1}{x}$$



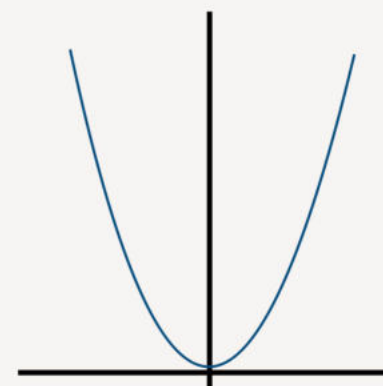
$$\bullet x \rightarrow \sqrt{x}$$



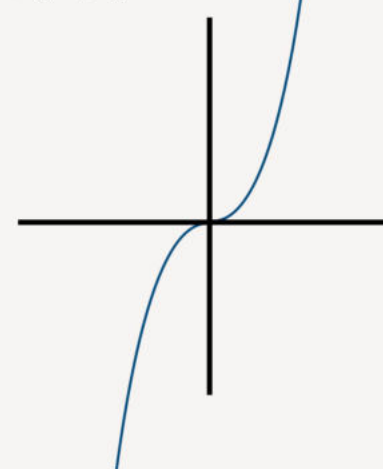
$$\bullet x \rightarrow |x|$$



$$\bullet x \rightarrow x^2$$



$$\bullet x \rightarrow x^3$$





42 Rue Pierre Nicole
75005
Paris, France

+33 6 61 24 47 83
mimprep.com
contact@mimprep.com

Master Application
All Inclusive Support

Interview Coaching only

Free ressources